
Plan Overview

A Data Management Plan created using DMPonline

Title: Coherent states in numerical methods for fermionic particle-preserving quantum systems

Creator: Michal Horansky

Principal Investigator: Michal Horanský

Data Manager: Michal Horanský

Project Administrator: Dmitry Shalashilin

Affiliation: University of Leeds

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Template: University of Leeds: Data Management Plan

Project abstract:

We design a new array of numerical methods for electronic structure theory. These methods all utilise Gilmore-Perelomov coherent states. To recover the electronic ground state we sample these coherent states around the Hartree-Fock state using low-excitation-guided sampling; to extrapolate potential energy surfaces, we decompose a known FCI solution into these states using rejection sampling from the wavefunction. We focus on fine-tuning the sampling process in a way which is scalable with molecule size. We also investigate a special mathematical construction which approximates the coherent state construction and lowers the time complexity of matrix element calculations.

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Coherent states in numerical methods for fermionic particle-preserving quantum systems

Project Information

Researcher Name(s)

Michal Horanský

Faculty

- Faculty of Engineering and Physical Sciences

School or Service

School of Chemistry

KRISTAL Reference

Question not answered.

Supervisor Name(s)

Dmitry Shalashilin

Ethical Review Reference

Question not answered.

Last DMP Review Date

2026-07-29

Provide a brief overview of your project, including proposed research methods.

We want to design new methods for electronic structure theory. I mainly design and test new sampling methods on the coherent state manifold and then deploy them to the two main problems we study: estimating the electronic ground state and extrapolating the electronic potential energy surfaces. Every such test entails running a Monte Carlo algorithm on an array of molecular systems (taken from

the geometry parameter space), outputting the sampled coherent states, their effective Hamiltonian matrices, and the estimated energy eigenstates and eigenvalues. This is compared against HF and FCI methods to get a sense of method accuracy.

Key Questions

What data will be produced? Are you using data from other sources?

The computational part of the project entails running Monte Carlo calculations using the proposed methods on the investigated molecules. The outputs of these calculations will be the numerical results of the calculation.

So far, we do not use data from other sources. We might outsource some hard-to-calculate solutions for some larger molecules further down the line for the sake of benchmarking, but there is no concrete talk of this yet.

Where will data be stored? Include file formats and approximate volume.

The calculations are primarily run on HPC Aire under my account vzwt2244. The data is stored there and copied to my personal computer at regular intervals in order to be analysed and displayed.

The outputs are stored in .csv and .json files; a single calculation produces around 20MB of data. The entire volume of data we're expecting to work with should not exceed 10GB.

Who will have access to data during the project? Give details of collaborators and agreements.

During the project, only I have direct access to the data because my credentials are needed to access either my files on Aire or my personal computer. My supervisor can ask me to audit the files, and the Aire administrators can access the data stored on Aire. We do not share the data with anyone outside the research group during the project.

Ethics and legal compliance: Any ethical, contractual, special, sensitive or consent issues?

Given the nature of the data, there are no ethical or legal concerns connected to it.

How will data be documented and described?

I curate the data and periodically discard useless data which was produced by methods not promising to our goal. In the progress reports I produce, the most promising or relevant data is represented visually in graphs.

What are your training and support needs in relation to research data management?

I do not have any training or supporting needs in relation to research data management.

What are the plans for data sharing beyond project partners? How is this impacted if some of your data needs to be restricted?

We are planning on storing the data used in all of our published articles on a public github repository (together with the source code used) and attaching the access link to all the articles for the sake of academic integrity and assistance with reproduction. It is also possible that, if the method is successful, we might integrate some version of the source code in the COSMOS library for quantum chemistry. The repository will be distributed with the MIT license.

The data we produce and use is generated either by our own code or by other open-source code: no legal body can restrict access to it in the future.

Will any sensitive or commercially exploitable intellectual property be generated?

No, the data generated is meant to approximate known results for well-studied molecules. It is not benchmarked on industry-protected molecules or against industry-protected methods. While the ability to quickly analyse the properties of the electronic structure of a molecule may be used in a commercial setting, we do not have any reason to fear that making our scripts and data publicly available will in any way disrupt any part of modern chemical industry.

Who is responsible for managing the data? What resources will you need?

As the main collector and analyser of the data, I have responsibility for managing them. Primarily, I must be careful not to disrupt Aire by mismanaging the data stored there; I was given proper guidance on this by the official documentation released by the University and require no further resources.